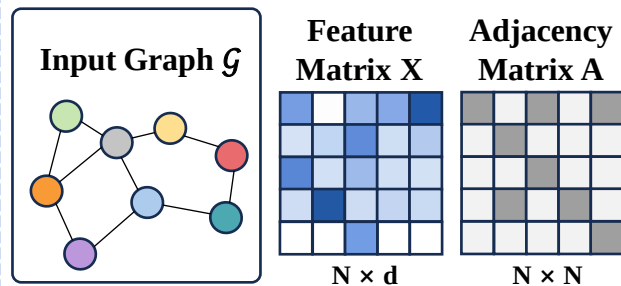


Pre-trained GNN: Spectral Bias

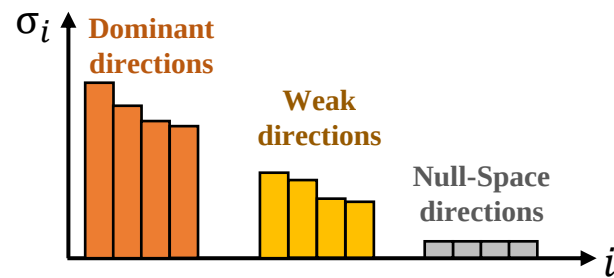


GNN Layer l

$$h_i^{(l-1)} \rightarrow W^{(l)} \rightarrow h_i^{(l)}$$

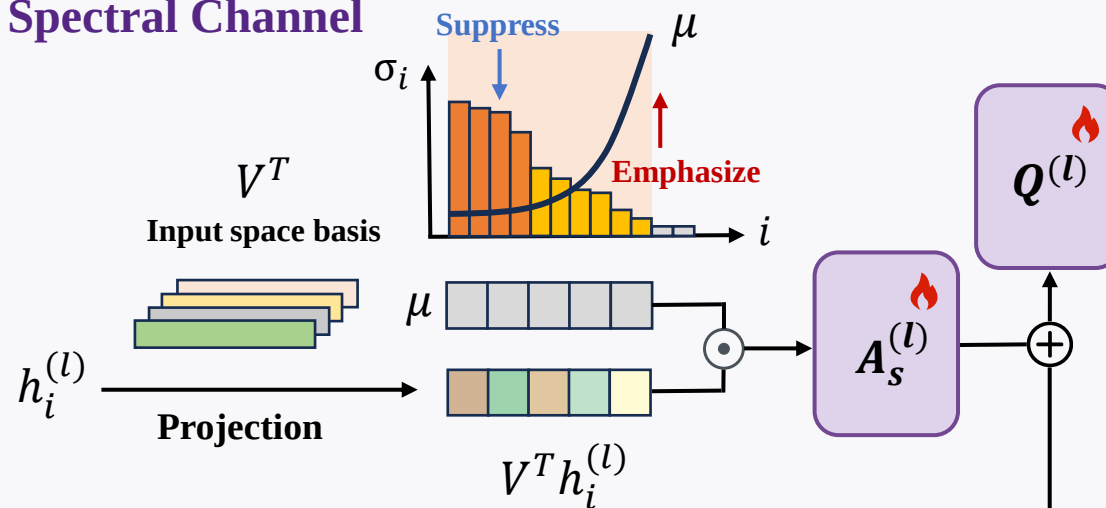
SVD decomposition $W = U \Sigma V^T$

Singular Value Spectrum (Σ)

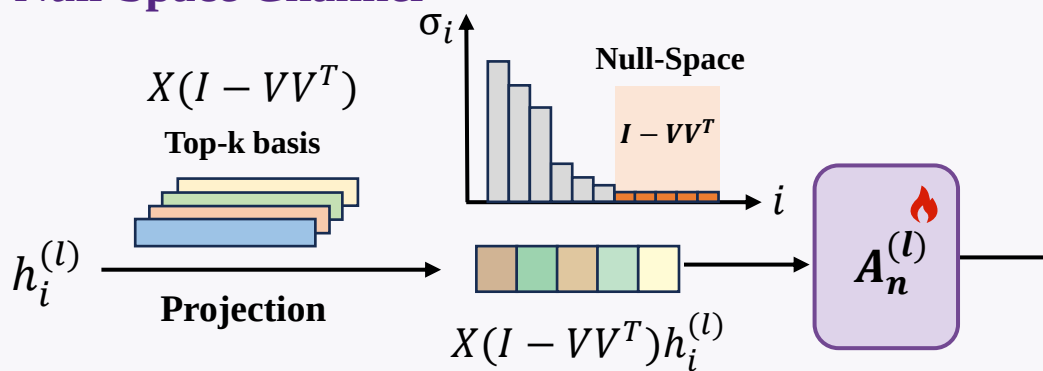


Spectral Reverse Prompt (SRP)

Spectral Channel

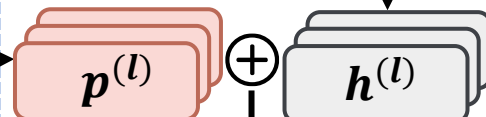


Null-Space Channel

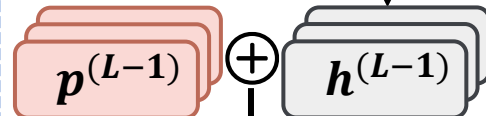


Prompt Tuning

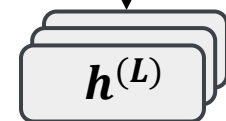
Layer l



...



Layer L



Downstream

🔥 Trainable ❄️ Frozen